By the end of these exercises, you will be able to:

✅ Use conditional statements (if, elif, else) to control the flow of a program.

✅ Apply the modulus operator (%) to check number properties like divisibility and parity.

✅ Implement input validation and string comparisons to build basic user authentication logic.

✅ Use a while loop to repeat tasks based on conditions such as success criteria or attempt limits.

✅ Combine loops and decision-making logic to build interactive programs like simple menus.

✅ Practice using a for loop with the range() function to iterate over numbers efficiently.

```
Start coding or generate with AI.
```

1) 🔐 Scenario-Based Exercise: Security Door Access

Scenario: You are a security engineer at a high-tech facility. To enter the building, employees must type in a secret code on a keypad. The code for today is "123". You need to write a program that checks if the person has entered the correct code.

🧠 Mission Brief: Your task is to write a program that:

Asks the person to enter the access code.

If the code is "123", the system should print: "🔓 Access granted. Welcome inside!"

If the code is incorrect, the system should print: "⛔ Access denied. Incorrect code."

```
Start coding or generate with AI.
```

```
secret_code=123
employee_code=int(input("Enter the code:"))
if secret_code==employee_code:
  print("Access granted.welcome inside!")
else:
  print("Access denied.Incorrect code")
```

```
Enter the code:678
Access denied.Incorrect code
```

2) You’re building a simple number-checking tool for a maths learning app. The app should tell users whether their chosen number is divisible by 3 and 7.

🧪 Task: Write a Python program that:

Asks the user to enter a number.

Checks whether the number is divisible by both 3 and 7.

Displays a clear message with the result.

```
user_number = int(input("Enter a number: "))
if user_number % 3 == 0 and user_number % 7 == 0:
    print("The number is divisible by both 3 and 7")
else:
    print("The number is not divisible by both 3 and 7")
```

```
Enter a number: 56
The number is not divisible by both 3 and 7
```

```
Start coding or generate with AI.
```

3) You're developing a simple calculator tool for a local community learning website. Users should be able to choose what operation they want to perform and continue using the calculator until they decide to exit.

🧪 Task: Write a Python program that:

Displays a menu with the following options:

- 1- Addition
- 2- Subtraction
- 3- Multiplication

- 4-Division
- 5- Exit

Asks the user to choose an operation

Asks for two numbers, then performs the chosen operation

Repeats the menu until the user selects Exit

```
#simple calculator
while True:
    print("1-Addition")
    print("2-Subtraction")
    print("3-Multiplication")
    print("4-Division")
    print("5-Exit")
    choice = int(input("Enter your choice: "))

    if choice == 1:
        num1 = int(input("Enter your first number: "))
        num2 = int(input("Enter your second number: "))
        print(num1 + num2)
    elif choice == 2:
        num1 = int(input("Enter your first number: "))
        num2 = int(input("Enter your second number: "))
        print(num1 - num2)
    elif choice == 3:
        num1 = int(input("Enter your first number: "))
        num2 = int(input("Enter your second number: "))
        print(num1 * num2)
    elif choice == 4:
        num1 = int(input("Enter your first number: "))
        num2 = int(input("Enter your second number: "))
        print(num1 / num2)
    elif choice == 5:
        break
    else:
        print("Invalid choice")
```

```
1-Addition
2-Subtraction
3-Multiplication
4-Division
5-Exit
```

```
Start coding or generate with AI.
```

4) You’re building a fitness tracker app. The app should calculate the total number of steps someone would take if they increase their steps by 1 each day — starting from 1 up to a number they choose.

🍼 Task: Write a Python program that:

Asks the user to enter a positive number

Calculates the sum of all numbers from 1 to that number

Displays the total

If the user enters 10, the output should be:

"The sum of numbers from 1 to 10 is 55"

```
Ask the user for a positive number
num=int(input("Enter a positive number:"))
Calculate the sum from 1 to that number
total=0
for i in range(1,num+1):
total=total+i
Display the total
print(f"The sum of numbers from 1 to {num} is {total}")
```

```

```

5) You've been asked to help design a feature for a Math Quiz App. One challenge asks the user to enter a number — and your program will check whether it's a prime number (a number that has only two factors: 1 and itself).

🪄 Task: Write a Python program that:

Asks the user to enter a positive integer

Checks whether the number is a prime number

Displays a message telling the user whether the number is prime or not

```
user_number=int(input("Enter a positive number:"))
is_prime=True
if user_number<2:
  is_prime=False
else:
  for i in range(2,user_number):
      if user_number %i==0
      is_prime=False
      break
if is_prime:
  print(f"{user_number} is a prime number")
else:
  print(f"{user_number} is not a prime number")
```

```
Start coding or generate with AI.
```

6) You're building a tool for a puzzle game where certain levels require calculating the factorial of a number to unlock a clue. A factorial of a number is the product of all positive integers from 1 up to that number.

For example: 5! = 5 × 4 × 3 × 2 × 1 = 120

🪄 Task: Write a Python program that:

Asks the user to enter a non-negative integer

Calculates the factorial of that number

Displays the result

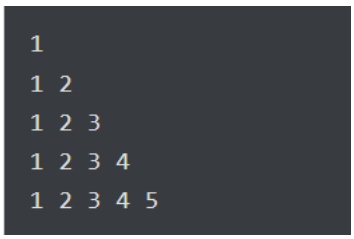
```
# Ask user for a non-negative integer
num=int(input("Enter a non-negative integer:"))
# Calculate the factorial
factorial = 1
for i in range(1, num +1):
  factorial=factorial*1
# Display the result
print(f"The factorial of {num} is {factorial}")
```

```
Start coding or generate with AI.
```

7) You're helping design a study tool for kids that displays fun number patterns. One such pattern looks like a staircase, where each line adds one more number.

🪄 Task: Write a Python program using nested for loops to display the following pattern:

Hint: Use nested for loops




```
# outer loop (1 to 5)
for i in range(1 to 6):
  # inner loop (1-i)
  for j in range(1,i+1):
```

```
    print(j,end=" ")
  print()
```

```
# Go through numbers from 1 to 100
for i in range(1, 101):
  if i % 3==0 and i % 5 ==0
  print ("Fizzbuzz")
  elif i % 3==0:
  print("Fizz")
  elif i % 5==0:
  print("Buzz")
  else:
  print(i)
```

Start coding or generate with AI.